

Jahid Md Hasan

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SUMMARY

AI Engineer focused on Generative AI, Agentic workflows, and scalable AI systems. Experienced in building end-to-end AI pipelines, RAG applications, and production-oriented AI solutions across healthcare, finance, and cybersecurity domains. Skilled in Python, FastAPI, LangChain, vector databases, and cloud-based AI workflows with experience in automation, deployment pipelines, monitoring, evaluation, and modular AI system development.

WORK EXPERIENCE

Machine Learning Intern - Unified Mentor

June 2025 - Sep 2025

- Developed end-to-end machine learning pipelines across healthcare, finance, and accessibility domains, covering data preprocessing, feature engineering, model training, and deployment workflows.
- Built a fraud detection system using temporal transaction data and domain-specific simulation rules, leveraging feature engineering, imbalanced classification techniques, and ensemble models (XGBoost, Random Forests).
- Engineered a healthcare diagnostic model for heart disease prediction, incorporating explainable AI tools (SHAP/LIME) and performance metrics (ROC-AUC, precision-recall) for clinical interpretability.

Data Science Intern - InLighnX Global

Oct 2025 - Feb 2026

- Built an Air Quality Prediction System using Random Forest, Prophet, and LSTM, achieving $R^2 = 0.9968$, with advanced feature engineering, automated preprocessing, and interactive forecasting dashboard deployment.
- Developed an AI-powered voice interaction system integrating speech recognition, NLP-based command routing, and event-driven automation workflows.
- Applied end-to-end ML lifecycle practices including data cleaning, feature engineering, model evaluation, serialization, and deployment-ready packaging.

PROJECTS

DocuMind AI — Agentic RAG Document Assistant

[View in GitHub](#)

- Built an end-to-end Retrieval-Augmented Generation (RAG) system with an agentic router using LLM function-calling (Groq / Llama 3.1 8B) to dynamically select between document retrieval, summarization, and math tools based on user intent, eliminating hardcoded pipelines..
- Engineered a PDF ingestion pipeline using PyMuPDF and sentence-transformers (all-MiniLM-L6-v2) with 500-token overlapping chunks persisted to a local ChromaDB vector store, achieving sub-200ms retrieval on CPU.
- Deployed an interactive multi-turn Q&A interface via Gradio with conversation memory and source citations, supporting both cloud (Groq API) and fully local inference (Ollama + Phi-3 mini) modes.

Network Security for Phishing Detection

[View in GitHub](#)

- Designed a **production-ready ML pipeline** utilising an artefact-driven approach to manage the full lifecycle, including ETL, ingestion, and transformation, ensuring the workflow is scalable and observable.
- Built a **robust monitoring system** using the KS-test for data drift detection and schema validation to maintain training consistency and prevent silent model degradation in live environments.

Real-Time Face Mask Detection System

[View in GitHub](#)

- Developed a modular, multi-stage OpenCV system to track forklift laser perimeters, delivering sub-millisecond latencies and deterministic execution entirely on CPU-bound edge nodes.
- Combined union-find (DSU) clustering and minAreaRect hulls to handle perspective warps, integrating a state-machine tracker with polygon IoU to suppress 62% of jitter.

EDUCATION

Amity University – Noida

Master of Computer Applications (MCA) – Specialization in AR/VR & Intelligent Systems

Sep 2023 – March 2026

Assam University – Silchar

Bachelor of Science (BSc.) – Computer Science Honours

Aug 2019 – Nov 2022

SKILLS

Programming/APIs: Python, FastAPI, SQL, REST APIs, Git
GenAI & Agents: LangChain, RAG, Agentic Workflows, Tool Calling, MCP, Prompt Engineering, AI Evaluation
LLM & Retrieval: Transformers, Hugging Face, Embeddings, Semantic Search, Vector Databases (FAISS, Chroma)
AI/ML: PyTorch, TensorFlow, Scikit-learn, Model Evaluation, Experimentation
Cloud & MLOps: AWS, Docker, MLflow, CI/CD, Drift Detection
Data Engineering: ETL Pipelines, MongoDB Atlas, Data Validation
Certifications & Training: IBM Data Science | TCS AI Pro | IBM RAG & Agentic AI | RoboAI Industrial Training